

Key Changes

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This document lists changes made to the Model Context Protocol (MCP) specification since the previous revision, 2025-11-25.

1. Major changes

1. Remove protocol-level sessions and the `Mcp-Session-Id` header from the Streamable HTTP transport. List endpoints (`tools/list` , `resources/list` , `prompts/list`) no longer vary per-connection. Servers that need cross-call state use explicit, server-minted handles passed as ordinary tool arguments (SEP-2567).
2. Make MCP stateless: remove the `initialize` / `notifications/initialized` handshake. Every request now carries its protocol version and client capabilities in `_meta` (`io.modelcontextprotocol/protocolVersion` , `io.modelcontextprotocol/clientCapabilities`). Clients SHOULD identify themselves on each request (`io.modelcontextprotocol/clientInfo`), and servers SHOULD identify themselves in each result's `_meta` (`io.modelcontextprotocol/serverInfo`). Version mismatches return `UnsupportedProtocolVersionError` (SEP-2575).
3. Add `server/discover` : servers MUST implement this RPC to advertise their supported protocol versions, capabilities, and identity. Clients MAY call it before any other request for up-front version selection, or use it as a backward-compatibility probe on STDIO (SEP-2575).
4. Replace the HTTP GET endpoint and `resources/subscribe` / `resources/unsubscribe` with `subscriptions/listen` : a single long-lived POST-response stream for opted-in server-to-client change notifications. Clients opt in to specific types (`toolsListChanged` , `promptsListChanged` , `resourcesListChanged` ,

([SEP-2575](#)).

5. Remove `ping` , `logging/setLevel` , and `notifications/roots/list_changed` . Log level is now set per-request via `io.modelcontextprotocol/logLevel` in `_meta` ; servers MUST NOT emit `notifications/message` for requests that did not include this field ([SEP-2575](#)).
6. Move experimental tasks out of the core protocol and into an official extension (`io.modelcontextprotocol/tasks`). The redesigned extension replaces the blocking `tasks/result` method with polling via `tasks/get` and a new `tasks/update` for client-to-server input, removes `tasks/list` , and allows servers to return task handles unsolicited without per-request opt-in ([SEP-2663](#)).
7. Multi Round-Trip Requests (MRTR) pattern introduced which replaces the previous approach of sending server-initiated requests, such as `roots/list` , `sampling/createMessage` , or `elicitation/create` . Servers return an `InputRequiredResult` (`resultType: "input_required"`) whose `inputRequests` field carries the requests for the additional information needed to process the request. Clients respond with `inputResponses` on a retry of the original request providing the requested information. ([SEP-2322](#)).
8. All results now carry a required `resultType` field: `"complete"` for ordinary results and `"input_required"` for multi round-trip request interim results. Clients **MUST** treat results from earlier-protocol servers that omit the field as `"complete"` ([SEP-2322](#)).
9. Remove SSE stream resumability and message redelivery (the `Last-Event-ID` header and SSE event IDs) from the Streamable HTTP transport. A broken response stream loses the in-flight request; clients **MUST** re-issue it as a new request with a new request ID ([SEP-2575](#)).

2. Minor changes

3. Servers **SHOULD** return tools from `tools/list` in a deterministic order to enable client-side caching and improve LLM prompt cache hit rates.
4. Require standard MCP request headers (`Mcp-Method` , `Mcp-Name`) on Streamable HTTP POST requests, and add support for custom headers from tool parameters via `x-mcp-header` ([SEP-2243](#)).
5. Require `ttlMs` and `cacheScope` fields on results returned by `tools/list` , `prompts/list` , `resources/list` , `resources/read` , and `resources/templates/list` via a new `CacheableResult` interface. `ttlMs` is a freshness hint (in milliseconds) allowing clients to cache responses and reduce polling; `cacheScope` (`"public"` or `"private"`) controls whether shared intermediaries may cache the response. Both fields complement existing `listChanged` notifications ([SEP-2549](#)).
6. Change resource not found error code from `-32002` to `-32602` (Invalid Params) to align with JSON-RPC specification.
7. Authorization servers **SHOULD** include the `iss` parameter in authorization responses per [RFC 9207](#), and MCP clients **MUST** validate a present `iss` against the recorded issuer before redeeming the authorization code ([SEP-2468](#)).
8. Require MCP clients to specify an appropriate `application_type` during Dynamic Client Registration to avoid OpenID Connect redirect URI conflicts ([SEP-837](#)).
9. Clarify that client credentials are bound to the authorization server that issued them: clients **MUST** key persisted credentials by the issuer identifier, **MUST NOT** reuse them with a different authorization server, and **MUST** re-register when the authorization server changes ([SEP-2352](#)).
10. Loosen `inputSchema` and `outputSchema` to allow any JSON Schema 2020-12 keywords, and `structuredContent` to allow any JSON value. Add `$ref` resolution requirements and composition-keyword resource bounds ([SEP-2106](#)).
11. Remove the `notifications/elicitation/complete` notification and the `elicitationId` field of URL mode elicitation requests, both introduced in `2025-11-`

`requestState` .

- Define an **error code allocation policy** partitioning the JSON-RPC server-error range:
 - `-32000` to `-32019` remains implementation-defined (existing SDK usage is grandfathered), `-32020` to `-32099` is reserved for the MCP specification. Renumber the error codes introduced in this draft accordingly — `HeaderMismatch` `-32001` → `-32020` , `MissingRequiredClientCapability` `-32003` → `-32021` , `UnsupportedProtocolVersion` `-32004` → `-32022` — and add `HeaderMismatchError` to the schema, which previously existed only in transport prose.

3. Deprecated

Features listed here remain part of the specification but are scheduled for removal under the **feature lifecycle and deprecation policy**. New implementations should not adopt them. The **deprecated features registry** tracks every feature currently in the Deprecated state.

- Deprecate the Roots, Sampling, and Logging features (**SEP-2577**). These features remain fully functional during the deprecation window but new implementations should not add support for them. Suggested migrations: pass directories or files via tool parameters, resource URLs, or server configuration instead of Roots; integrate directly with LLM provider APIs instead of Sampling; log to `stderr` (stdio) or use OpenTelemetry instead of Logging.
- Reclassify the HTTP+SSE transport (deprecated since protocol version `2025-03-26`) as Deprecated under the feature lifecycle policy (**SEP-2596**). Migrate to **Streamable HTTP**.
- Reclassify the `includeContext` values `"thisServer"` and `"allServers"` (soft-deprecated since protocol version `2025-11-25`) as Deprecated (**SEP-2596**). Omit the field or use `"none"` ; these values will be removed no later than the Sampling feature itself.

4. Other schema changes

1. `schema.json` now correctly reflects that the Typescript definition of minimum/maximum/default are `number` 's and not just `integers` . This was caused by running the generator using `--defaultNumberType integer` ([PR#2710](#)).

5. Governance and process updates

1. Adopt a specification [feature lifecycle and deprecation policy](#) defining the Active, Deprecated, and Removed feature states, a minimum twelve-month deprecation window, and a [registry of deprecated features](#) ([SEP-2596](#)).

6. Process changes

1. Formalize PR-based SEP workflow with markdown files in `seps/` directory, PR-derived numbering, sponsor responsibilities, and status management via PR labels ([SEP-1850](#)).

7. Full changelog

For a complete list of all changes that have been made since the last protocol revision, [see GitHub](#).

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